

Adaptive Kernel Selection for Stein Variational Gradient Descent

Anonymous Authors

Abstract

Efficiently approximating posterior distributions is a central challenge in Bayesian inference. Stein Variational Gradient Descent (SVGD) is a leading variational inference method that transports a set of particles to approximate a target distribution. The SVGD dynamics are governed by a reproducing kernel Hilbert space (RKHS) and depend critically on the choice of the kernel function, which directly determines both convergence and approximation quality. The commonly used median heuristic is simple to implement but rigid, and it fails in high-dimensional settings. We resolve this limitation with a principled strategy for adaptively choosing kernel parameters over an abstract family of kernels. Recent convergence analyses based on the kernelized Stein discrepancy (KSD) establish that optimizing the kernel parameters by maximizing the KSD improves performance. Building directly on this insight, we introduce Adaptive SVGD (Ad-SVGD), a method that alternates between updating the particles via SVGD and adaptively tuning kernel bandwidths through gradient ascent on the KSD. We give a theoretical analysis that extends existing results on minimizing the KSD for fixed kernels to our adaptive setting, proving convergence of the maximal KSD over our kernel class. Our experiments confirm this: Ad-SVGD consistently outperforms standard heuristics across a range of tasks.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle-based method for approximate Bayesian inference that has established itself as a compelling alternative to traditional Markov Chain Monte Carlo (MCMC) methods. SVGD evolves a set of particles using update directions derived from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with updates constrained to lie within the unit ball of a reproducing kernel Hilbert space (RKHS). The decisive limitation of SVGD is its sensitivity to kernel choice, which governs the algorithm’s performance [8, 28]. Compounding this, the resulting particle approximation systematically underestimates the posterior variance [2]. These observations have hardened into a widely held belief that SVGD fails as the dimension of the underlying state space increases. We overturn this belief: by introducing an adaptive mechanism that dynamically tunes the kernel during inference through maximization of the kernelized Stein discrepancy (KSD), we enable effective transport in complex and high-dimensional spaces.

1.1 Related work.

Since its introduction [18], SVGD has become a standard tool for approximate Bayesian inference across a wide range of machine learning applications [22, 24, 30, 15]. Recent work has substantially advanced the theoretical understanding of SVGD. Mean-field convergence has been established in both continuous-time [23, 8, 5] and discrete-time [16, 32] settings, and finite-particle convergence

rates have been proven under various assumptions [3, 33]. The SVGD dynamics have further been identified as gradient flows on probability distribution spaces [19, 8], in direct analogy to Jordan, Kinderlehrer, and Otto [14].

The performance of SVGD hinges on the choice of the kernel function: it dictates the interaction between particles and the overall convergence of the method (see also Figures 1 and 9). Convergence results typically concern mean-field convergence with respect to KSD, whose relation to weak convergence is itself determined by the selected kernel [11]. The commonly used median heuristic [12] is simple to implement but has no theoretical justification and demonstrably degrades as the dimensionality of the problem increases [2, 40, 36]. A line of recent work has sought to mitigate this high-dimensional degradation [7, 37, 10, 21], and others have modified the algorithm by using neural networks to learn the update direction [17, 39].

The approach most closely related to ours is Ai et al. [1], which introduces a mixture-of-kernels framework. Their method defines a KSD for a weighted linear combination of kernels and learns the kernel weights by maximizing this multiple-kernel KSD. Their approach, however, is confined to finite kernel bases and never exploits the continuous parameter optimization that we develop here.

1.2 Contributions.

In this work, we settle a fundamental research question: whether SVGD, without any architectural or algorithmic enhancements, achieves competitive performance when equipped with a principled and adaptively chosen kernel. By isolating kernel selection from the other modifications found in SVGD variants, we pinpoint the extent to which kernel choice alone governs the effectiveness of SVGD, and we show how far adaptive kernel learning pushes the capabilities of the original method.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We introduce a novel method that dynamically updates the kernel parameters by maximizing the KSD during SVGD inference. Unlike the commonly used median heuristic, which is restricted to a single scalar bandwidth, our approach optimizes multiple continuous kernel parameters, delivering markedly greater flexibility and adaptivity throughout the SVGD updates.
- (ii) **Theoretical Analysis.** We ground our algorithm theoretically by analyzing it in the discrete-time mean-field setting, extending existing convergence results for SVGD with fixed kernels. Specifically, we prove that the supremum of the KSD over a parameterized kernel class converges to zero as the particle distribution approaches the target. Assuming a Stein logarithmic Sobolev inequality, we further derive iteration complexity in the mean-field limit.
- (iii) **Empirical Validation.** Through numerical experiments, we demonstrate that our adaptive kernel selection consistently outperforms the median heuristic and substantially alleviates variance collapse.

2 Mathematical background

We begin by considering a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and its associated RKHS $\mathcal{H}_0$. We define $\mathcal{H}$ as the d -fold Cartesian product of $\mathcal{H}_0$ equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. Moreover, we denote

by $x \cdot y$ the Euclidean inner product and $\nabla \cdot$ the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted by $\mathcal{P}(\mathbb{R}^d)$ and $\mathcal{P}_p(\mathbb{R}^d)$ denotes the subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, we define the Wasserstein p -distance

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify the distance between two measures through the maximum deviation of integrals over a class of test functions. To make this approach tractable for measuring the distance of a sample to an intractable target distribution, Stein's method [35] constructs test functions with zero mean w.r.t. the target. Indeed, for the operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable choice of kernel, we have $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. They defined the *kernelized Stein discrepancy* (KSD) as

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. This optimization problem is solved by $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, and Φ_k is the feature map associated with the kernel k . As a result, the supremum evaluates to the RKHS norm of ψ , giving the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and reference distribution μ_0 , SVGD transforms μ_0 into an approximation of π by choosing $\mu_{n+1} := T_\# \mu_n$, where $T_\#$ is the push-forward operator for the map $T(x) = x + \gamma \psi^{\mu_n}(x)$, with the vector field $\psi^\mu = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ being the direction of steepest descent. This is motivated by the fact that the solution of Equation (1) implies that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ is the minimizer of $\left. \frac{d}{d\gamma} \text{KL}(T_\# \mu_n \parallel \pi) \right|_{\gamma=0}$ in the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \parallel \cdot)$ denotes the KL-divergence. In particular, for this choice we have

$$\left. \frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_\# \mu_n \parallel \pi \right) \right|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Iteratively applying the maps T generated in this way to a particle set $\{X_0^i\}_{i=1}^M$ sampled from μ_0 leads to the following particle updates:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because proving convergence of SVGD with respect to the KL divergence is difficult and demands restrictive assumptions, recent work has turned to analyzing convergence in terms of the kernelized Stein discrepancy (KSD) [16, 33, 32]. Minimizing the KSD alone, however, does not guarantee weak convergence: the sequence of mean-field measures may fail to be tight, a necessary condition for weak convergence of probability measures in Polish spaces. Moreover, the optimization problem in Equation (1) makes clear that the RKHS structure—and therefore the choice of kernel—directly determines how well convergence in KSD translates into weak convergence [11].

We resolve this issue and strengthen convergence guarantees by formulating our adaptive variant of SVGD over a parameterized family of kernels $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, we denote by KSD_θ the corresponding *kernelized Stein discrepancy*, by ψ_θ^μ the optimal update direction, and by Φ_θ the associated feature map. This explicit parameterization lets us adaptively select kernels during optimization, so that we directly exploit the kernel’s influence on convergence and remove the limitations of relying on a fixed kernel choice.

Our approach rests on a clear principle: it is advantageous to select the kernel that maximizes the KSD between the empirical particle distribution and the target distribution. The reasoning is direct. The instantaneous decrease in KL divergence under SVGD is proportional to the squared KSD at the current particle measure (see Equations (2) and (7)). Hence, at any iteration, choosing the kernel with the largest KSD maximizes the rate of KL decrease. Geometric analyses of SVGD as a gradient flow confirm this perspective, showing that kernels inducing larger KSD values yield more favorable convergence properties when comparing the associated RKHSs [28, 8].

Whereas vanilla SVGD only guarantees convergence of the KSD for a single fixed kernel, our algorithm targets the worst-case KSD across the entire kernel class. As opposed to the median heuristic used by Liu and Wang [18], the approach is applicable to any class of parameterized kernels: At each step of the algorithm, we try to find an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

We realize this idea by adjusting the kernel parameter with one or more gradient ascent steps on the KSD, using a step size $s > 0$, before executing the particle update; we also allow the parameter to be updated less frequently than every SVGD step to cut runtime (see Algorithm 1). This option is controlled by a user-specified decision variable `paramupdate`. Crucially, the base SVGD step for transporting the particles can be swapped for any more advanced variant (e.g., adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates) without touching our kernel-selection mechanism, and the gradient ascent on the kernel parameter can likewise be replaced by any alternative optimization scheme. Our implementation of the kernel update is based on the formula [6, 20]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the necessary gradient is computed directly. Two points establish that Ad-SVGD adds only marginal computational cost over vanilla SVGD:

- The gradient ascent steps for the kernel parameter reuse the same gradients $\nabla \log \pi(X_n^i)$ as the corresponding SVGD step, so Ad-SVGd requires no additional gradient evaluations of $\log \pi$.
- The remaining cost is easily controlled by using a small number of update steps nstepstheta for the kernel parameter, or by updating the kernel parameter only after multiple SVGD steps. When the number of particles M is large, subsampled particles suffice to empirically approximate the KSD.

Algorithm 1: Ad-SVGd

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps nsteps , $\text{nstepstheta} \in \mathbb{N}$

Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to  $\text{nsteps} - 1$  do
    Decide  $\text{paramupdate} \in \{\text{True}, \text{False}\}$ ;
    if  $\text{paramupdate}$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $\text{nstepstheta} - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2\left(\frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi\right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider a target measure with Lebesgue density of the form $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To firmly ground our proposed Ad-SVGd, we extend recent results on the convergence of SVGD in the sense of KSD. Concretely, we extend the convergence analysis of [32] under Talagrand's inequality, which holds under mild assumptions on the target distribution and is strictly weaker than the commonly employed logarithmic Sobolev inequality. We adopt the same assumptions as [32], imposed uniformly over all kernel parameters.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand's inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and make use of the following fundamental inequality. Given a fixed kernel parameter $\theta \in \Theta$ such that Assumption 3 is satisfied, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then, according to Proposition 3.1 in [32], it holds

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$ one can then derive the descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The key aspect to verify (7) is the verification of $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ using Talagrand's inequality, Assumption 2, which allows to apply (6) for $g = \psi_\theta^{\mu_n}$. Condition (7) can then be used to argue that

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

since $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) > 0$.

When introducing an adaptive kernel parameter choice (θ_n) , the inequality (7) changes to

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

Suppose the adaptive SVGd iteration can be written in the following simplified form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \arg\max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

meaning that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi).$$

This formulation is available whenever the maximization of the KSD with respect to the kernel parameter has a (unique) closed-form solution. This is precisely the case in the multiple-kernel SVGD framework [1], which parameterizes the kernel as a convex combination of base kernels

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For the general setting, the kernel parameter update θ_n in (9) needs to be approximated. Note that in the general setup we assume that $\text{argmax}_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition is a direct consequence of Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The situation changes when we assume that we can only approximately solve the maximization task of the KSD with respect to the kernel parameter. Suppose that $\max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n \end{aligned} \tag{10}$$

for some sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ with the goal to maximize $\text{KSD}_{\theta}(\mu_n | \pi)$. Specifically, in Algorithm 1 Ψ_n corresponds to the gradient ascent update for the KSD. However, as mentioned above, this update could be replaced by other suitable iterative optimization schemes. Our required assumption on the update rule is the following convergence behavior.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Using this assumption, we can make the following convergence guarantee.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under the so-called Stein logarithmic Sobolev inequality, we further sharpen the convergence of Ad-SVGD to convergence in the KL divergence. For specific scenarios of ε_n we derive explicit iteration complexity bounds for the mean-field limiting system (10). This setting moreover lets us characterize precisely the bias of Ad-SVGD when the KSD maximization is solved only up to a fixed $\bar{\varepsilon}$ -accuracy in each iteration. The details are given in Section B. Finally, we stress a key advantage of the adaptive setting: we require only a single kernel in our kernel class $\{k_{\theta}\}$ to satisfy the Stein logarithmic Sobolev inequality, which makes this condition far easier to verify.

5 Numerical Experiments

In this section, we evaluate Ad-SVGD on two numerical experiments: a one-dimensional Gaussian mixture model and a linear inverse problem governed by an ordinary differential equation. Additional experiments probing the effect of increasing dimensionality are provided in Section A. In Section 5.4 we further demonstrate the clear advantages of Ad-SVGD over SVGD with the median heuristic on a Bayesian logistic regression model, a benchmark established in prior SVGD literature.

5.1 Kernel parameterization

SVGD is most commonly used with kernels of the form

$$k_h(x, y) = \exp \left(-\frac{\|x - y\|_p^p}{h} \right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. We will focus on selection strategies for the parameter h , which is known as the kernel *bandwidth*. The commonly used heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med denotes the current median distance between the particles. This choice is motivated by the goal of balancing the two terms in the SVGD update (3) [18].

To exploit the flexibility of our adaptive method, we use product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, i.e. we allow for dimension-dependent bandwidths. The derivatives needed to apply Algorithm 1 with these kernels (see Equations (4) and (5)) are readily calculated. For comparison, we also tested an adjusted median heuristic with these kernels that takes a naive median for each dimension; this approach failed outright, collapsing the variance.

The flexibility of Ad-SVGD extends well beyond these experiments to more scalable kernel families in higher-dimensional settings. Spectral or low-rank kernel parametrizations, in particular, adapt the dominant directions of variability without incurring the exponential cost of full product kernels, while Matérn or mixed-product constructions supply additional control over smoothness and anisotropy.

5.2 Toy example

We first consider the one-dimensional example from Liu and Wang [18]: a Gaussian mixture with two components, $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We use $M \in \{50, 200, 500\}$ particles and sample the initial particle set from $\mathcal{N}(0, 1)$. We run SVGD for 10^4 steps with a step size of 1, using kernels of the form described above with $p = 1$ and different choices of (fixed) bandwidth h . As a measure of sample quality, we use the Wasserstein 1-distance $\mathcal{W}_1$, computed via the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) and an exact sample of size 10^5 . The results are shown in Figure 1: the algorithm performs well only for bandwidths within a narrow range. This sharp sensitivity to h makes a careful selection strategy indispensable.

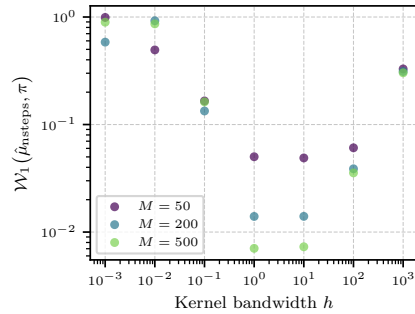


Figure 1: Final Wasserstein 1-distances for the one-dimensional example using SVGD with different fixed bandwidths h .

5.3 Linear inverse problem based on ODE

The next example is adapted from [38, Example 2.1]. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observation points of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. These observations are described by

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is defined by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ being the observation operator at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we consider a Gaussian process (GP) prior for u given by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$. The resulting problem is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ with prior given by $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For the implementation, we replace H by a numerical discretization operator for Equation (11) using a grid with mesh size 2^{-8} and consider the fully observed system (i.e. $N_{\text{obs}} = 2^8$). We use $N_x = 16$ terms for the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$. We compare the performance of SVGD using the median heuristic (which we call Med-SVGd) with our Ad-SVGd for particle ensemble sizes $M \in \{50, 100, 200\}$. We use the kernels described in Section 5.1 with $p = 1$. We run both algorithms for $4 \cdot 10^5$ iterations using the step size 10^{-3} for particle updates and, as suggested by Liu and Wang [18], a variant of AdaGrad for adaptive step size control. In Ad-SVGd we use the step size 10^{-5} for the bandwidth updates and update the bandwidth only once every 100 particle updates. Under this scheme, Med-SVGd and Ad-SVGd incur no significant runtime difference.

Figure 2 shows the GP reconstruction for an exemplary seed. Both methods approximate the mean well, but only Ad-SVGd correctly captures the posterior uncertainty. To quantify the approximation quality, we use the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the normal distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ is the sample mean and $\hat{\Sigma}$ the sample covariance of the particle set [29, Section 3.2]. Since the target π is a multivariate normal distribution, this admits the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the mean and covariance of the posterior. We also compare the marginal variances of the final particle distribution with the posterior. Figure 3 reports these results aggregated over 56 random seeds (note that the posterior covariance does not depend on $\bar{y}$). Ad-SVGd clearly achieves better approximations of the posterior than Med-SVGd, which underestimates the uncertainty of the problem. Moreover, and in sharp contrast to Med-SVGd, the approximation

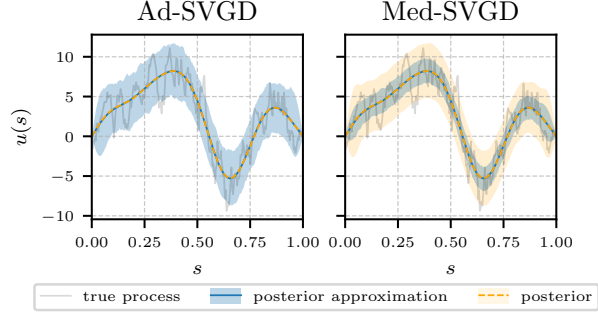
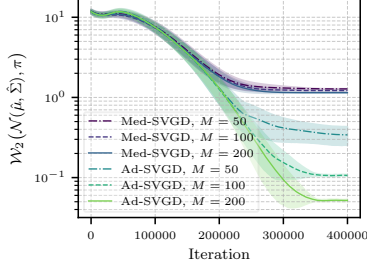
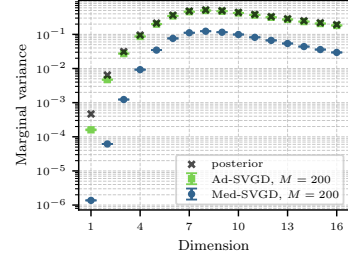


Figure 2: GP reconstruction for the ODE-based inverse problem, showing mean and 90% confidence interval

quality of Ad-SVGD improves as the number of particles increases beyond 50. Figure 4 shows the behavior of the bandwidths determined by Ad-SVGD. The component-wise bandwidths stabilize faster than the approximation error, and they exhibit clear differences across the components, with the final bandwidths negatively correlated with the corresponding marginal variances.

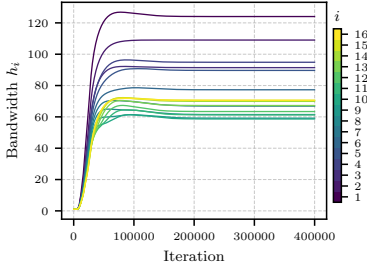


(a) Convergence of approximation error

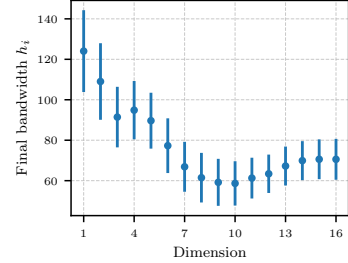


(b) Marginal variances

Figure 3: Aggregated results (mean and 95% confidence interval over 56 random seeds) for the ODE-based inverse problem using Med-SVGD and Ad-SVGD. Ad-SVGD attains lower approximation error and accurately recovers the marginal variances, whereas Med-SVGD underestimates them.



(a) Evolution of bandwidth per dimension (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Behavior of the bandwidth parameter for the ODE-based inverse problem using Ad-SVGD, aggregated over 56 random seeds

5.4 Bayesian logistic regression

As in Liu and Wang [18] and Liu et al. [21], we consider a Bayesian logistic regression (BLR) model applied to the Covtype data set [4]. For the regression weights w , we assign a Gaussian prior $w | \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, and we infer the posterior of $x = [w, \alpha]$. Following Liu et al. [21], we use the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] to generate a reference against which we evaluate the posterior approximation quality of Med-SVGD and Ad-SVGD. Again following Liu et al. [21], we test the methods on subsets of the original data set of size 1000 and aggregate our results over 10 random draws. To evaluate performance on the classification task, we compute the prediction accuracy over a test set using the particle mean. As Figure 5 shows, Ad-SVGD outperforms Med-SVGD across nearly all seeds, attaining higher test accuracy while sitting closer to the MCMC reference.

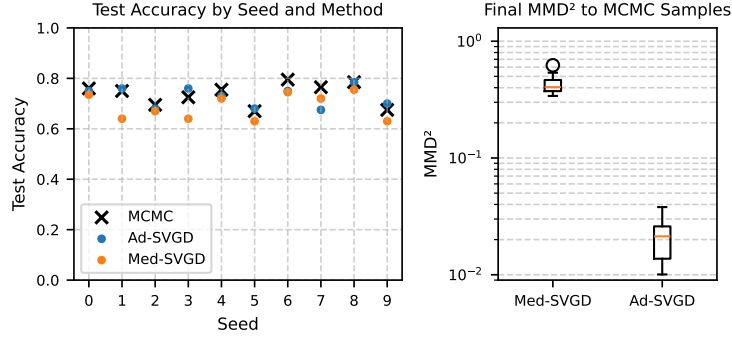


Figure 5: Approximation quality of SVGD particles for Bayesian logistic regression, measured by prediction accuracy (left) and MMD^2 to reference samples (right). Ad-SVGD achieves higher accuracy and lies closer to the MCMC reference.

To measure the accuracy of the posterior approximation, the same figure also reports the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples. The two methods differ markedly in how well they capture the posterior distribution. Figure 7 confirms this, showing the covariance matrices of the final particles compared to the MCMC reference, averaged over the 10 seeds (this visualization style follows Liu et al. [21]). Med-SVGD severely underestimates the posterior uncertainty, whereas Ad-SVGD delivers a substantially better approximation. Figure 6 further shows the evolution of the bandwidths for each component: from an uninformed initialization (1 in every component), Ad-SVGD reliably recovers suitable bandwidths that differ from the initialization by several orders of magnitude.

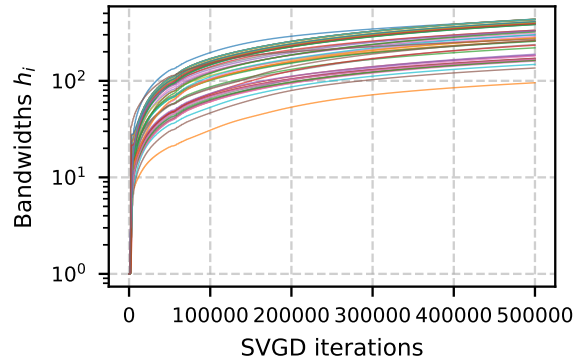


Figure 6: Bandwidth evolution for the BLR example

6 Limitations and outlook

The primary limitation of our analysis is its reliance on Assumption 4. Our ongoing work characterizes precisely when Assumption 4 is satisfied by the alternating gradient-ascent scheme in Algorithm 1 used in our experiments. While we do not yet prove that Assumption 4 holds in general, our implementation consistently produced strong empirical results.

Our analysis deliberately focused on the original SVGD dynamics, and a natural and promising direction is to combine the proposed adaptive kernel selection with recent variants such as sliced SVGD [10], Grassmann SVGD [21], or Stein transport [27], where our kernel-selection mechanism applies directly.

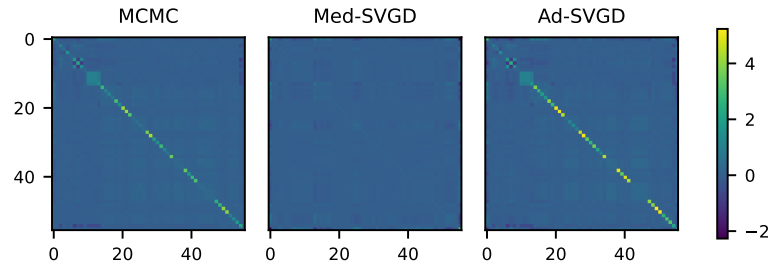


Figure 7: Covariance matrices of the final SVGD particle sets compared to MCMC samples (mean over 10 seeds). Ad-SVG recovers the reference covariance closely, while Med-SVG collapses it.

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A More numerical experiments

A.1 Gaussian mixture models

We revisit the example from Section 5.2. For the setup described there, Figure 9 shows histograms of the final particle distribution against the target density for $h \in \{0.001, 1, 1000\}$. We now compare Med-SVGD with Ad-SVGD, varying the number of particles M and comparing the final particle distributions after 10^4 iterations with step size 1.

Figure 8 shows the Wasserstein 1-distance between the final empirical distribution of the particles and the target distribution using Med-SVGD

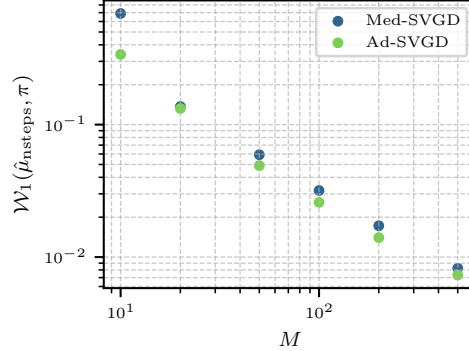


Figure 8: Approximation error for the 1D Gaussian mixture.

and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. As expected, the approximation quality improves with N for both methods, and both reach a Wasserstein distance below 0.01 for $N = 500$.

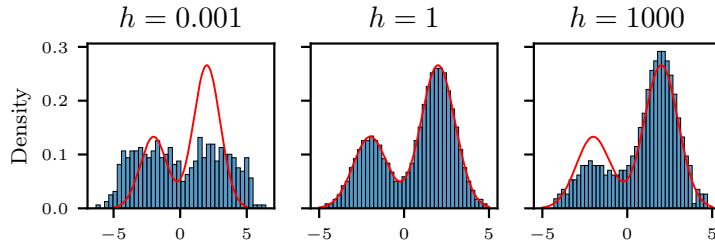


Figure 9: Histograms of the final particle set generated using fixed bandwidths $h = 0.001, 1, 1000$; target density π shown in red for comparison.

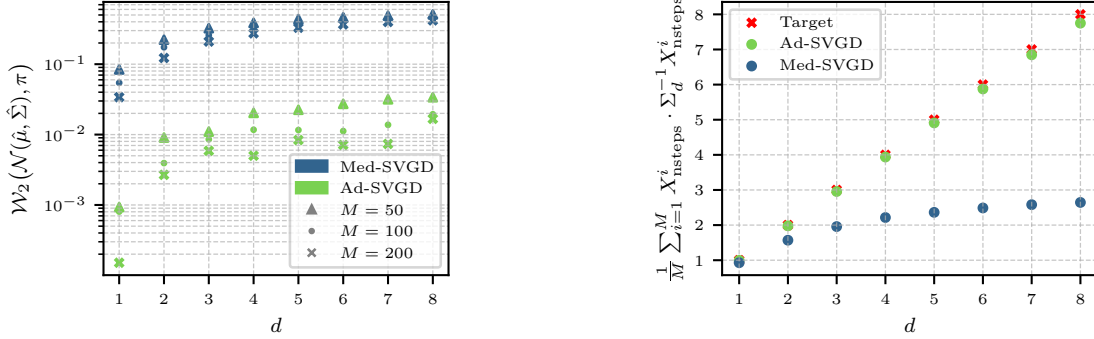
In the one-dimensional case, both methods approximate the target distribution well. Our adaptive bandwidth selection matches the median heuristic here; as we show next, its decisive advantage emerges in higher dimensions.

A.2 Scaling dimension: multivariate normal distribution

Moving into higher dimensions, we now consider the Gaussian target distributions $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. We used $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ as the initial particle distribution and ran the algorithms for 10^4 iterations with step size 0.1. Where necessary for numerical stability, we used a smaller step size and adjusted the number of iterations accordingly.

We compare the approximation quality of the final set of particles generated by Med-SVGD and Ad-SVGD as the dimensionality of the problem increases. Because the Wasserstein 2-distance between two Gaussian distributions has an explicit formula (see [29]), we use as our measure of sample quality the Wasserstein 2-distance between the target distribution π_d and the Gaussian

distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance matrix computed from the set of particles [29, Section 3.2]. Following the corresponding function of the Python Optimal Transport library [9] used for our calculations, we call this the *Bures Wasserstein distance*. Figure 10a tracks this sample quality measure for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as the dimensionality of the problem increases. Ad-SVGD decisively outperforms Med-SVGD for every value of d .



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal distributions of increasing dimension. Ad-SVGD maintains accurate approximations across all dimensions, while Med-SVGD degrades sharply.

For the Bures Wasserstein distance, the smaller variances in the last components of our target distributions π_d carry little weight; to expose the effect of failing to capture the uncertainty in these components, we consider the test statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom (i.e. it has expected value d) for $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). We compute the mean of this statistic on the set of particles with $M = 200$ and compare it to the expected value d in Figure 10b. Beyond the one-dimensional case, the test statistic for Med-SVGD drifts ever further from the true expected value as d increases, whereas the statistic for Ad-SVGD stays close to the true expectation. Med-SVGD thus fails to approximate higher-dimensional distributions, while Ad-SVGD handles these examples accurately.

We give further details on the final particle distributions for $M = 200$ and various d in Tables 1 and 2, which list, for each d , the marginal variances of the set of particles alongside the marginal variances of the target distribution (given in the first row). Table 1 reports Med-SVGD and Table 2 reports Ad-SVGD. Ad-SVGD accurately recovers the target distribution in all components in terms of the marginal variances, whereas the particles generated by Med-SVGD severely underestimate the uncertainty of the target distribution—exactly as Figure 10b already indicated.

Lastly, we focus on the marginal particle distributions for $M = 200$ and $d = 8$. To aid visualization, we normalized them by scaling each component of the particles with the inverse of the corresponding marginal standard deviation of the target distribution (i.e. we multiplied the i -th component by i), which turns each marginal distribution of π_d into a standard normal distribution. Figures 11 and 12 show the histograms of these normalized marginal particle distributions for $d = 8$ generated by Med-SVGD and Ad-SVGD, respectively, with a standard Gaussian density shown in each plot for comparison. Once again, Ad-SVGD captures all marginal distributions well, while Med-SVGD underestimates the uncertainty of the target distribution. The same conclusion is evident in Figures 13 and 14, where the quantiles of the normalized marginal particle distributions are compared

Table 1: Marginal variances of the final particle distribution for the d -dimensional examples generated by Med-SVGD with $M = 200$; marginal variances of the target distribution π_d shown for comparison. Med-SVGD markedly underestimates the variances, worsening as d grows.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final particle distribution for the d -dimensional examples generated by Ad-SVGD with $M = 200$; marginal variances of the target distribution π_d shown for comparison. Ad-SVGD recovers the target variances accurately across all components and dimensions.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

against the target quantiles (i.e. against a standard normal distribution).

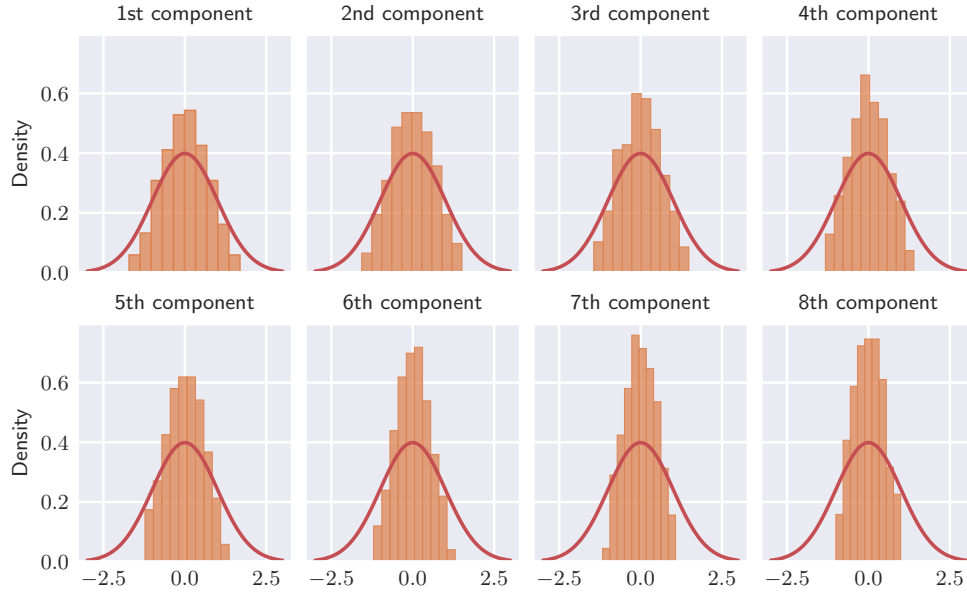


Figure 11: Histograms of the single components of the final set of particles for the eight-dimensional example generated by Med-SVGD with $M = 200$ and rescaled using the marginal target variances; standard Gaussian density shown for comparison. The marginals are visibly too concentrated.

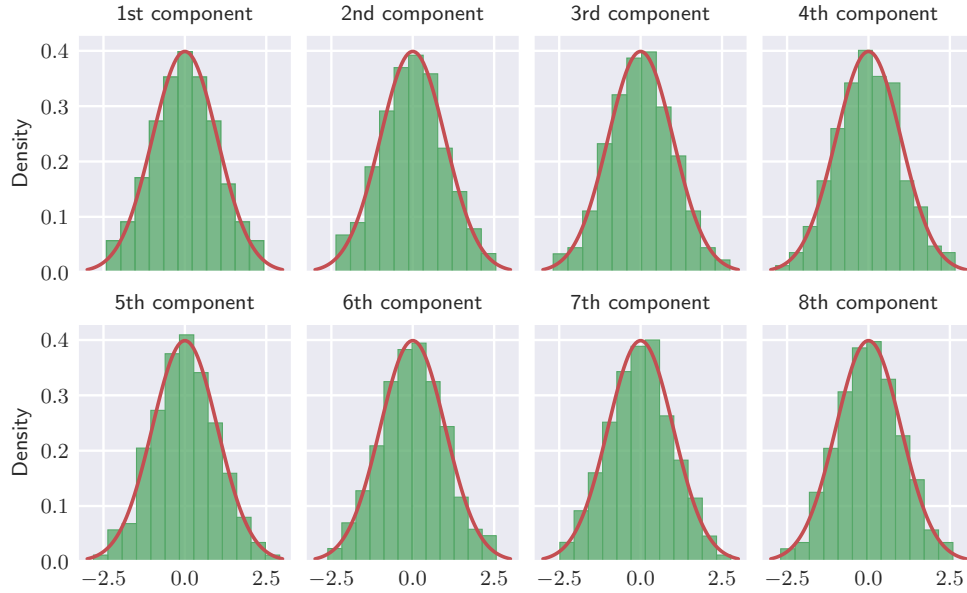


Figure 12: Histograms of the single components of the final set of particles for the eight-dimensional example generated by Ad-SVGD with $M = 200$ and rescaled using the marginal target variances; standard Gaussian density shown for comparison. Every marginal closely matches the standard Gaussian.

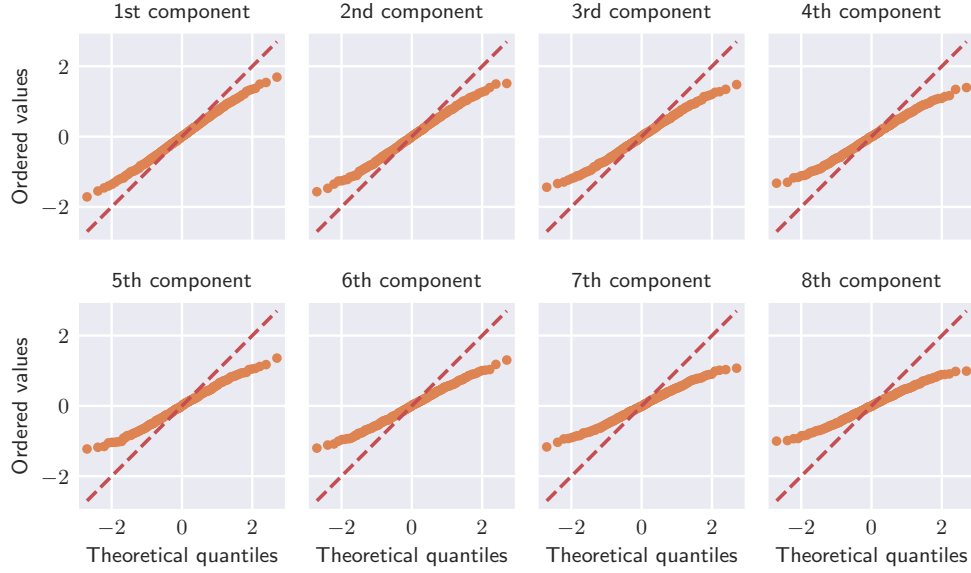


Figure 13: Q-Q plots comparing the marginals of the final particle distribution for the eight-dimensional example generated by Med-SVGd with $M = 200$ and rescaled using the marginal target variances against a standard normal distribution; line of slope 1 through the origin shown for comparison. The marginals deviate systematically from the reference line.

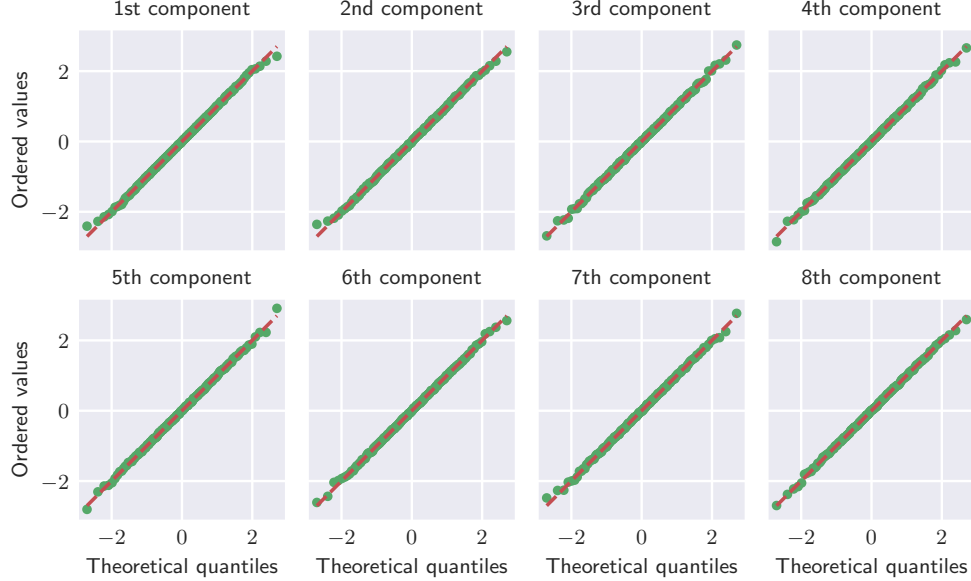


Figure 14: Q-Q plots comparing the marginals of the final particle distribution for the eight-dimensional example generated by Ad-SVGd with $M = 200$ and rescaled using the marginal target variances against a standard normal distribution; line of slope 1 through the origin shown for comparison. The marginals track the reference line closely.

A.3 Gaussian process inference

We further demonstrate the behavior of our Ad-SVGD under scaling dimensions on a Gaussian process (GP) inference task proposed in [31]. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ is a vector of independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this corresponds to an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal matrix Σ with entries k^{-2} , $k = 1, \dots, N_x$ and we assume independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_y}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

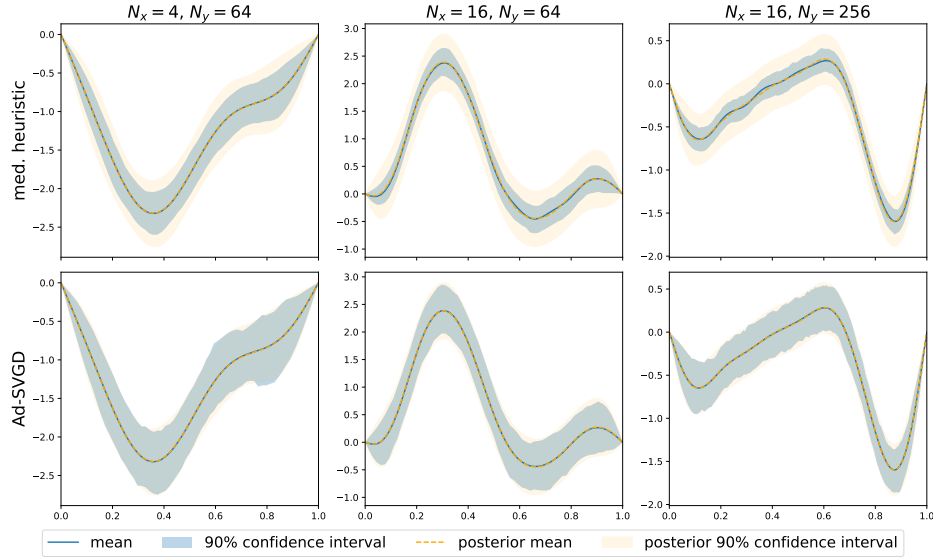


Figure 15: Estimated processes generated by the median heuristic and Ad-SVGD with $M = 100$ particles for different model configurations, compared to the posterior. Ad-SVGD tracks the posterior variance across configurations, while the median heuristic collapses it.

We use SVGD to sample from the resulting posterior $X|Y = \bar{y}$ and compare the median heuristic with our adaptive approach. For SVGD with the median heuristic, we use kernels of the form $k(x, y) = \exp(-\|x - y\|_1/h)$; for Ad-SVGD, we use product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, i.e. a different bandwidth for each dimension. Following [18], we use a variant of Adagrad for step size control in the scenarios with $N_x = 16$.

Figure 15 shows the resulting processes against the posterior for the choices $N_x = 4$ and $N_y = 64$, $N_x = 16$ and $N_y = 64$, as well as $N_x = 16$ and $N_y = 256$. As the dimension increases, SVGD with the median heuristic underestimates the posterior variance, while Ad-SVGD delivers a markedly better approximation—and does so consistently across the different numbers of observations of the Gaussian process. Table 3 quantifies this by comparing the trace of the covariance of the particle distributions generated by SVGD with the true posterior: SVGD with the median heuristic severely underestimates the uncertainty, whereas Ad-SVGD captures the variance far more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the covariance of the final particle distribution ($M = 100$ particles) compared to the theoretical posterior for different configurations, averaged over 25 runs. Ad-SVGD closely matches the theoretical trace, while the median heuristic falls well short.

B Refined theoretical analysis

We can sharpen our convergence result under a so-called Stein log-Sobolev inequality, which relates the KL divergence to the KSD. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed conditions under which it holds. A key benefit of our adaptive setting is that we relax this condition substantially: we require only a single kernel in the entire kernel class to satisfy the log-Sobolev inequality. More precisely, we make the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Using this property, we quantify convergence directly in terms of the KL divergence. Notably, this scenario yields an explicit error bound on $\mathcal{F}(\mu_n)$ even under the approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, without requiring $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□